

The critical $q = 1$ threshold for a full derivative of velocity averaging is sharp

Candidate counterexample for arXiv:1206.6659

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Status. Candidate full counterexample, likely valid; human review is needed.

1 Source question

Arsénio and Masmoudi prove a full gain of $1 + b - a$ derivatives at the critical index

$$\beta + D(1 - 1/p) = 1$$

when the Besov summation exponent is $q = 1$. In Theorem 3.2 they obtain only $s - \varepsilon$ when $q \neq 1$, and immediately ask whether the full gain is nevertheless possible [1].

$$(3.13) \quad \left\| \int_{\mathbb{R}^D} f(x, v) dv \right\|_{B_{p,q}^{s-\varepsilon}(dx)} \leq C \left(\|f\|_{B_{1,p,q}^{a,a}(\mathbb{R}_x^D \times \mathbb{R}_v^D)} + \|g\|_{B_{1,p,q}^{b,\beta}(\mathbb{R}_x^D \times \mathbb{R}_v^D)} \right),$$

where the constant $C > 0$ only depends on fixed parameters, in particular on $\varepsilon > 0$.

The remarks formulated above about Theorem 3.1 are still valid here regarding Theorem 3.2.

Thus, as long as $\beta + D(1 - \frac{1}{p}) < 1$, the above theorem yields a gain of regularity of $(1 + b - a) \frac{\alpha + D(1 - \frac{1}{p})}{1 + \alpha - \beta}$ (compared to the Sobolev embedding). Notice that this gain approaches a derivative of order $(1 + b - a)$ as β tends to $1 - D(1 - \frac{1}{p})$, or as α tends to infinity, which is optimal for a differential operator of order one. Therefore, the averaging lemma saturates beyond the value $\beta + D(1 - \frac{1}{p}) = 1$ and only yields at most a gain of $1 + b - a$ derivatives.

We do not know whether it is possible to achieve a full gain of $1 + b - a$ derivatives in the case $\beta + D(1 - \frac{1}{p}) = 1$ and $q \neq 1$. Nevertheless, the cases $\beta +$

Figure 1: The open critical-summation question in [1], PDF page 8.

The following construction answers the question negatively for every $q \in (1, \infty]$. It uses the admissible special case

$$D = p = 1, \quad \beta = 1, \quad a = b = 0, \quad \alpha = \frac{1}{2}. \quad (1.1)$$

For these parameters, a full gain would assert

$$\left. \begin{array}{l} f \in B_{1,1,q}^{0,1/2}(\mathbb{R}_x \times \mathbb{R}_v), \\ g = v \partial_x f \in B_{1,1,q}^{0,1}(\mathbb{R}_x \times \mathbb{R}_v) \end{array} \right\} \implies \rho(x) := \int_{\mathbb{R}} f(x, v) dv \in B_{1,q}^1(\mathbb{R}_x). \quad (1.2)$$

The mixed spaces and Littlewood–Paley blocks below are those defined in the source paper.

2 Counterexample theorem

Theorem 1. *For every $q \in (1, \infty]$ there are real tempered distributions f, g , supported in a fixed compact velocity interval, such that*

$$f \in B_{1,1,q}^{0,1/2}, \quad g = v\partial_x f \in B_{1,1,q}^{0,1}, \quad (2.1)$$

but

$$\int_{\mathbb{R}} f(x, v) dv \notin B_{1,q}^1(\mathbb{R}). \quad (2.2)$$

The velocity average is nevertheless an L^1 function. Hence (1.2) is false, and the critical $q = 1$ hypothesis in the source’s full-gain theorem is sharp within this Besov scale.

3 Proof intuition

The obstruction is exactly the endpoint sequence-space sum visible in the source proof. At spatial frequency λ_m , put m shrinking velocity atoms into f . Multiplication by v gives the atom at scale 2^{-j} an extra factor 2^{-j} ; after one velocity derivative this makes the $B_{1,q}^1$ transport cost of the m atoms comparable to their ℓ^q coefficient norm. Normalizing by $m^{-1/q}$ keeps that cost constant.

Velocity averaging behaves differently: all atoms have the same nonzero integral and therefore add in ℓ^1 . Their average has size $m m^{-1/q} = m^{1/q'}$. A final spatial coefficient sequence is in ℓ^q , so both hypotheses hold, but multiplication by $m^{1/q'}$ takes it out of ℓ^q . When $q = 1$, $q' = \infty$ and the bad factor disappears, exactly matching the positive theorem.

4 Velocity atoms

Choose $0 \leq \psi \in C_c^\infty((1, 3/2))$ with

$$c_\psi := \int_{\mathbb{R}} \psi(v) dv > 0,$$

and set, for $j \geq 1$,

$$\psi_j(v) = 2^j \psi(2^j v), \quad a_j(v) = v \psi_j(v) = a(2^j v), \quad a(w) = w \psi(w). \quad (4.1)$$

Thus

$$\int \psi_j = c_\psi, \quad \|a_j\|_{L^1} = 2^{-j} \|a\|_{L^1}. \quad (4.2)$$

Lemma 2 (Critical atomic estimate). *For $1 < q \leq \infty$ and every finite scalar sequence (c_j) ,*

$$\left\| \sum_j c_j a_j \right\|_{B_{1,q}^1(\mathbb{R})} \leq C_q \|(c_j)\|_{\ell^q}. \quad (4.3)$$

Moreover,

$$\|\psi_j\|_{B_{1,q}^{1/2}} \leq C_q 2^{j/2}. \quad (4.4)$$

Proof. Let Δ_ℓ denote a velocity block at frequency 2^ℓ . For $\ell \leq j$, L^1 boundedness of the block gives

$$2^\ell \|\Delta_\ell a_j\|_{L^1} \leq C 2^{-(j-\ell)}. \quad (4.5)$$

For $\ell > j$, use an integer $M \geq 3$, the vanishing moments of the high-frequency kernel, and $\|\partial^M a_j\|_{L^1} = C_M 2^{j(M-1)}$ to obtain

$$2^\ell \|\Delta_\ell a_j\|_{L^1} \leq C_M 2^{-(M-1)(\ell-j)}. \quad (4.6)$$

The two right sides form an ℓ^1 convolution kernel in $\ell - j$. Young's sequence inequality proves the high-block part of (4.3). Its low block is bounded by $C \sum_j |c_j| 2^{-j} \leq C_q \|c\|_{\ell^q}$. Finally (4.4) is the standard scaling estimate (or follows from the same two-case block calculation) for $2^j \psi(2^j \cdot)$. $\square$

5 Construction and input norms

Choose a real Schwartz function H whose Fourier transform is supported in a narrow annulus on which one fixed Littlewood–Paley multiplier equals one. Put

$$k_m = 4m, \quad \lambda_m = 2^{k_m}, \quad H_m(x) = \lambda_m H(\lambda_m x). \quad (5.1)$$

The selected spatial annuli are disjoint, $\|H_m\|_{L^1} = \|H\|_{L^1}$, and $\|H'_m\|_{L^1} = \lambda_m \|H'\|_{L^1}$.

For finite q , use the usual conjugate exponent q' ; for $q = \infty$ set $1/q = 0$ and $1/q' = 1$. Define

$$F_m(v) = \lambda_m^{-1} m^{-1/q} \sum_{j=1}^m \psi_j(v). \quad (5.2)$$

Let (d_m) be specified below and set

$$f(x, v) = \sum_{m \geq 1} d_m H_m(x) F_m(v), \quad (5.3)$$

$$g(x, v) = \sum_{m \geq 1} d_m H'_m(x) v F_m(v). \quad (5.4)$$

The series converge in $\mathcal{S}'$ (and by norm for finite q), and termwise differentiation gives $v \partial_x f = g$.

By (4.4) and the triangle inequality,

$$\|F_m\|_{B_{1,q}^{1/2}} \leq C \lambda_m^{-1} m^{-1/q} \sum_{j=1}^m 2^{j/2} \leq C \lambda_m^{-1} m^{-1/q} 2^{m/2}. \quad (5.5)$$

Because $\lambda_m = 2^{4m}$, the right side decays exponentially. Finite overlap of the spatial blocks therefore gives

$$\|f\|_{B_{1,1,q}^{0,1/2}} < \infty \quad (5.6)$$

for every polynomially bounded (d_m) .

On the other hand, Lemma 2 gives

$$\|v F_m\|_{B_{1,q}^1} \leq C \lambda_m^{-1} m^{-1/q} \|(1, \dots, 1)\|_{\ell^q} = C \lambda_m^{-1}. \quad (5.7)$$

Combining this with $\|H'_m\|_1 = C \lambda_m$ and the separated spatial annuli yields

$$\|g\|_{B_{1,1,q}^{0,1}} \leq C \|(d_m)\|_{\ell^q}, \quad (5.8)$$

with the evident supremum interpretation when $q = \infty$.

6 The average fails at exactly one derivative

Every atom in (5.2) has the same integral, and hence

$$\int_{\mathbb{R}} F_m(v) dv = c_\psi \lambda_m^{-1} m^{1/q'}. \quad (6.1)$$

Therefore

$$\rho(x) = c_\psi \sum_{m \geq 1} d_m \lambda_m^{-1} m^{1/q'} H_m(x). \quad (6.2)$$

The multiplier selected in (5.1) is one on the m th Fourier support. Thus, for $q < \infty$,

$$\|\rho\|_{B_{1,q}^1}^q \geq c \sum_{m \geq 1} |d_m|^q m^{q/q'}, \quad (6.3)$$

and for $q = \infty$ the right side is replaced by $c \sup_m |d_m| m$.

If $1 < q < \infty$, take

$$r = \frac{1 + 1/q}{2}, \quad d_m = m^{-r}. \quad (6.4)$$

Then $rq = (q + 1)/2 > 1$, so $(d_m) \in \ell^q$ and (5.8) is finite. But (6.3) contains

$$\sum_{m \geq 1} m^{q-1-rq} = \sum_{m \geq 1} m^{(q-3)/2}, \quad (6.5)$$

which diverges for every $q > 1$. If $q = \infty$, take $d_m = 1$; (5.8) is finite while $\sup_m m = \infty$. This proves (2.1)–(2.2).

Finally, (6.2) converges absolutely in L^1 because $\lambda_m^{-1} m^{1/q'} |d_m|$ is summable. The velocity supports in (4.1) are all contained in a fixed compact interval. This completes the proof of Theorem 1. $\square$

7 Verification, scope, and novelty

The parameter check is exact:

$$\alpha = \frac{1}{2} > 0 = -D(1 - 1/p), \quad b = 0 \geq a - 1 = -1, \quad \beta + D(1 - 1/p) = 1.$$

The construction fails to contradict the known $q = 1$ result for exactly the right reason: then $m^{1/q'} = 1$, so an ℓ^1 input sequence cannot make the output sum diverge. The nonessential script included in the packet checks finite partial sums of (5.8) and (6.5) for representative q .

Cheap run indexes and bounded searches through 12 August 2026 used the arXiv id, exact title, quoted question, “critical velocity averaging”, “full gain”, “Besov q ”, “sharpness”, and “counterexample”. They found the source/published paper and later work in other velocity-averaging regimes, but no later paper explicitly resolving this exact critical mixed-Besov summation question. The proof is self-contained beyond the source’s standard Littlewood–Paley definitions.

8 Human-review recommendation

Verify (4.5)–(4.6) and the finite-overlap passage from the tensor products to the mixed Besov norms in (5.6)–(5.8). Those are the only substantive bookkeeping points. Once accepted, the packet gives a full negative answer to the source question for every $q > 1$, and establishes sharpness of $q = 1$.

References

- [1] D. Arsénio and N. Masmoudi, *A new approach to velocity averaging lemmas in Besov spaces*, J. Math. Pures Appl. 101 (2014), 495–551; arXiv:1206.6659.